

## Practical-1 (Java Basics)

**A. Write a java program that takes a number as input and prints its multiplication table upto 10.**

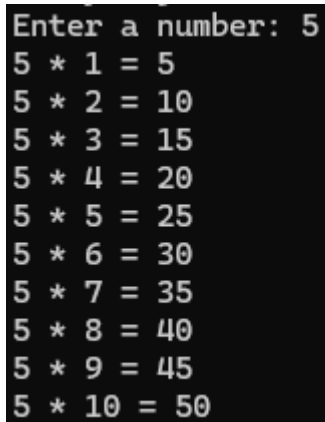
**Source Code:**

```
import java.util.Scanner;

public class Table {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = scanner.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(num + " * " + i + " = " + (num * i));
        }
    }
}
```

**Output:**



```
Enter a number: 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

**B. Write a java program to display the following pattern.**

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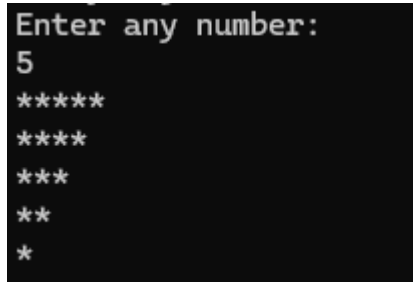
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**Source Code:**

```
import java.util.*;
public class Pattern {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter any number:");
        int n = sc.nextInt();
        for (int i = n; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

**Output:**

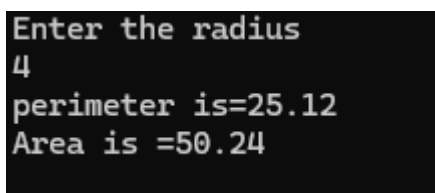


```
Enter any number:
5
*****
****
***
**
*
```

**C. Write a java program to find area and perimeter of a circle.****Source Code:**

```
import java.util.*;

public class Area {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the radius ");
        int radius = sc.nextInt();
        double perimeter = 2 * 3.14 * radius;
        double area = 3.14 * radius * radius;
        System.out.println("perimeter is=" + perimeter);
        System.out.println("Area is =" + area);
    }
}
```

**Output:**A screenshot of a terminal window showing the output of the Java program. The text is as follows:

```
Enter the radius
4
perimeter is=25.12
Area is =50.24
```

## Practical-2

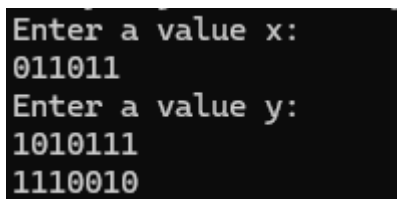
### (Use of Operator)

**A. Write a java program to add binary number.**

**Source Code:**

```
import java.util.*;
public class AddBinary {
static String add_Binary(String x, String y) {
int num1 = Integer.parseInt(x, 2);
int num2 = Integer.parseInt(y, 2);
int sum = num1 + num2;
String result = Integer.toBinaryString(sum);
return result;
}
public static void main(String args[]) {
Scanner sc = new Scanner(System.in);
System.out.println("Enter a value x:");
String x = sc.nextLine();
System.out.println("Enter a value y:");
String y = sc.nextLine();
System.out.print(add_Binary(x, y));
}
}
```

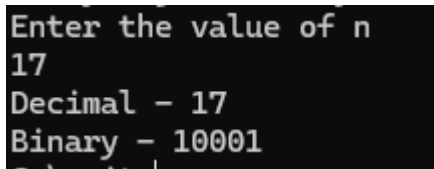
**Output:**



```
Enter a value x:
011011
Enter a value y:
1010111
1110010
```

**B. Write a java program to convert decimal to binary and vice versa.****Source Code:**

```
import java.util.*;
class binary {
static void decToBinary(int n)
{
int[] binaryNum = new int[1000];
int i = 0;
while (n > 0)
{
binaryNum[i] = n % 2;
n = n / 2;
i++;
}
for (int j = i - 1; j >= 0; j--)
System.out.print(binaryNum[j]);
}
public static void main (String[] args)
{
Scanner sc= new Scanner (System.in);
System.out.println("Enter the value of n ");
int n=sc.nextInt();
System.out.println("Decimal - " + n);
System.out.print("Binary - ");
decToBinary(n);
}
```

A screenshot of a terminal window showing the output of the Java program. The text is as follows:

```
Enter the value of n
17
Decimal - 17
Binary - 10001
```

**C. Write a java program to reverse a string.****Source Code:**

```
import java.io.*;
import java.util.Scanner;

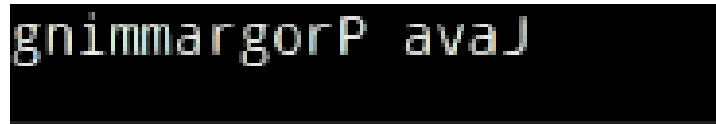
class Reverse {
    public static void main(String[] args) {

        String s = "Java Programming";
        String r = "";
        char ch;

        for (int i = 0; i < s.length(); i++) {

            ch = s.charAt(i);
            r = ch + r;
        }

        System.out.println(r);
    }
}
```

**Output:**

### Practical-3 (Java Data Type)

**A. Write a java Program to count the letters ,spaces, numbers and other character of an input.**

**Source Code:**

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the String:");
        String test = sc.nextLine();

        char[] ch = test.toCharArray();
        int letter = 0;
        int space = 0;
        int num = 0;
        int otherchat = 0;
        for(int i = 0; i < test.length(); i++){
            if(Character.isLetter(ch[i])){
                letter ++ ;
            }
            else if(Character.isDigit(ch[i])){
                num ++ ;
            }
            else if(Character.isSpaceChar(ch[i])){
                space ++ ;
            }
            else{
                otherchat ++;
            }
        }
        System.out.println("The string is : @sateeshm.com 123456");
        System.out.println("letter: " + letter);
        System.out.println("space: " + space);
        System.out.println("number: " + num);
        System.out.println("other: " + otherchat);

    }
}
```

**Output:**

```
Enter the String:
@java Programming 8
The string is : @sateeshm.com 123456
letter: 15
space: 2
number: 1
other: 1
```

**B.Implementing a java function that calculate the sum of digits for the given char array consisting of the digits 0 to 9 . the function should return the digit sum as a long value.**

**Source Code:**

```
public class Sum {  
    public static long calculateDigitSum(char[] digits) {  
        long sum = 0;  
        for (char digit : digits) {  
            if (digit >= '0' && digit <= '9') {  
                sum += (digit - '0');  
            } else {  
                throw new IllegalArgumentException("Invalid character in input array: " + digit);  
            }  
        }  
        return sum;  
    }  
    public static void main(String[] args) {  
        char[] digitArray = {'1', '2', '3', '4', '9'};  
        long result = calculateDigitSum(digitArray);  
        System.out.println("The sum of the digits is: " + result);  
    }  
}
```

**Output:**

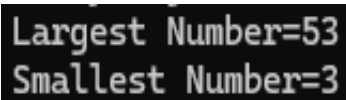


```
The sum of the digits is: 19
```



**C. Find smallest and largest element from the array.****Source Code:**

```
import java.util.*;
class Number{
public static void main(String args[]){
int arr[] = {12,3,53,21,18};
int largest = arr[0], smallest = arr[0];
for(int i=0; i<arr.length; i++){
if(smallest > arr[i])
    smallest = arr[i];
if(largest < arr[i])
    largest = arr[i];
}
System.out.println("Largest Number=" +largest);
System.out.println("Smallest Number=" +smallest);
}
}
```

**Output:**

```
Largest Number=53
Smallest Number=3
```

**A. Designed a class Sort that contains the method asec() and desc().****Source Code:**

```
import java.util.Arrays;

public class Sort {
    private int[] data;

    public Sort(int[] data) {
        this.data = data;
    }

    public int[] asec() {
        int[] sortedData = Arrays.copyOf(data, data.length);
        Arrays.sort(sortedData);
        return sortedData;
    }

    public int[] desc() {
        int[] sortedData = Arrays.copyOf(data, data.length);
        Arrays.sort(sortedData);
        for (int i = 0; i < sortedData.length / 2; i++) {
            int temp = sortedData[i];
            sortedData[i] = sortedData[sortedData.length - 1 - i];
            sortedData[sortedData.length - 1 - i] = temp;
        }
        return sortedData;
    }

    public static void main(String[] args) {
        int[] sampleData = {3, 1, 4, 1, 5, 9, 2};
        Sort sorter = new Sort(sampleData);

        System.out.println("Original Array: " + Arrays.toString(sorter.data));

        int[] ascendingSorted = sorter.asec();
        System.out.println("Sorted in Ascending Order: " + Arrays.toString(ascendingSorted));

        int[] descendingSorted = sorter.desc();
        System.out.println("Sorted in Descending Order: " + Arrays.toString(descendingSorted));
    }
}
```

**Output:**

```
Original Array: [3, 1, 4, 1, 5, 9, 2]
Sorted in Ascending Order: [1, 1, 2, 3, 4, 5, 9]
Sorted in Descending Order: [9, 5, 4, 3, 2, 1, 1]
```

**B. Design a class that demonstrate the use of constructor and destructor.****Source Code:**

```
class Student{
    int id;
    String name;
    Student(int i,String n){
        id = i;
        name = n;
    }

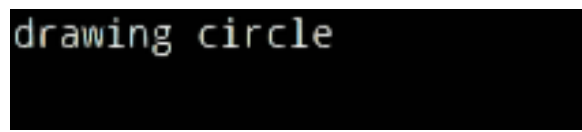
    void display(){System.out.println(id+" "+name);}
    public static void main(String args[]){
        Student s1 = new Student(111,"Karan");
        Student s2 = new Student(222,"Aryan");
        s1.display();
        s2.display();
    }
}
```

**Output:**

```
111 Karan
222 Aryan
```

**C. Write a java program to demonstrate the implementation of abstract class.****Source Code:**

```
abstract class Shape {  
    abstract void draw();  
}  
  
class Rectangle extends Shape {  
    void draw() {  
        System.out.println("drawing rectangle");  
    }  
}  
class Circle1 extends Shape {  
    void draw() {  
        System.out.println("drawing circle");  
    }  
}  
class TestAbstraction1 {  
    public static void main(String args[]) {  
        Shape s = new Circle1();  
        s.draw();  
    }  
}
```

**Output:**A screenshot of a terminal window with a black background. The text "drawing circle" is displayed in a light blue, monospaced font.

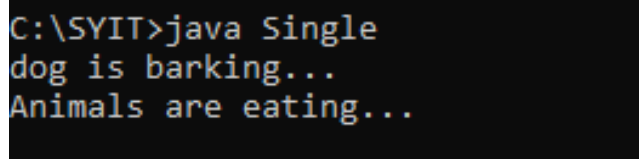
## Practical-5 (Inheritance)

**A. Write a java program to implement single inheritance.**

**Source Code:**

```
class Animal{
void eat(){
System.out.println("Animals are eating...");
}
}
class Dog extends Animal{
void bark(){
System.out.println("dog is barking...");
}
}
class Single{
public static void main(String args[]){
Dog d=new Dog();
d.bark();
d.eat();
}
}
```

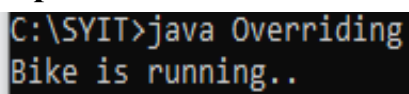
**Output:**



```
C:\SYIT>java Single
dog is barking...
Animals are eating...
```

**B. Write a java program to implement Method Overriding.****Source Code:**

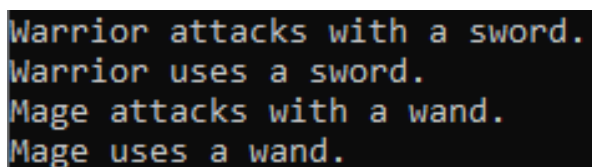
```
class Vehicle{
    void run(){
        System.out.println("Vehicle is running");
    }
}
class Bike extends Vehicle{
    void run(){
        System.out .println("Bike is running..");
    }
}
public class Overriding{
    public static void main(String args[]){
        Bike obj = new Bike();
        obj.run();
    }
}
```

**Output:**

```
C:\SYIT>java Overriding
Bike is running..
```

**C. Write a java Program to implement multiple inheritance .****Source Code:**

```
interface Weapon {  
    void use();  
}  
  
class Warrior implements Character, Weapon {  
    public void attack() {  
        System.out.println("Warrior attacks with a sword.");  
    }  
  
    public void use() {  
        System.out.println("Warrior uses a sword.");  
    }  
}  
  
class Mage implements Character, Weapon {  
    public void attack() {  
        System.out.println("Mage attacks with a wand.");  
    }  
  
    public void use() {  
        System.out.println("Mage uses a wand.");  
    }  
}  
  
public class Multiple {  
    public static void main(String[] args) {  
        Warrior warrior = new Warrior();  
        Mage mage = new Mage();  
  
        warrior.attack();  
        warrior.use();  
  
        mage.attack();  
        mage.use();  
    }  
}
```

**Output:**

```
Warrior attacks with a sword.  
Warrior uses a sword.  
Mage attacks with a wand.  
Mage uses a wand.
```

**Practical - 6**

**b. Write a java program to add two matrices and print the resultant matrix.**

**Source Code:**

```
public class MAddition {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int rows = input.nextInt();
        System.out.print("Enter the number of columns: ");
        int columns = input.nextInt();

        int[][] matrix1 = new int[rows][columns];
        int[][] matrix2 = new int[rows][columns];
        int[][] sum = new int[rows][columns];

        System.out.println("Enter elements for the first matrix:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix1[i][j] = input.nextInt();
            }
        }
        System.out.println("Enter elements for the second matrix:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix2[i][j] = input.nextInt();
            }
        }

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                sum[i][j] = matrix1[i][j] + matrix2[i][j];
            }
        }
        System.out.println("Sum of the matrices:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                System.out.print(sum[i][j] + " ");
            }
            System.out.println();
        }

        input.close();
    }
}
```

**Output:**



```
C:\syit>java MAddition
Enter the number of rows: 2
Enter the number of columns: 2
Enter elements for the first matrix:
3 4 7 3
Enter elements for the second matrix:
1 2 4 2
Sum of the matrices:
4 6
11 5
```

**b. Write a java program to accept element of 3\*3 matrices and develop new transpose matrices from it.**

### Source Code:

```
import java.util.Scanner;

public class Transpose {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int[][] matrix = new int[3][3];
        int[][] transpose = new int[3][3];

        System.out.println("Enter elements of the 3x3 matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrix[i][j] = scanner.nextInt();
            }
        }

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                transpose[j][i] = matrix[i][j];
            }
        }

        System.out.println("Original Matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }

        System.out.println("Transpose Matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(transpose[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

```
    scanner.close();  
}
```

**Output:**

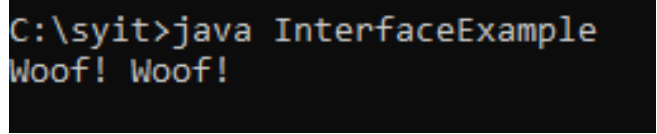
```
C:\syit>java Transpose  
Enter elements of the 3x3 matrix:  
1 2 3 4 5 6 7 8 9  
Original Matrix:  
1 2 3  
4 5 6  
7 8 9  
Transpose Matrix:  
1 4 7  
2 5 8  
3 6 9
```

**Practical - 7**

**A. Write a java code to implement interface for a defined class.**

**Source Code:**

```
interface Animal {  
    void makeSound();  
}  
  
class Dog implements Animal {  
    public void makeSound() {  
        System.out.println("Woof! Woof!");  
    }  
}  
  
public class InterfaceExample {  
    public static void main(String[] args) {  
        Dog myDog = new Dog();  
        myDog.makeSound();  
    }  
}
```

**Output:**

```
C:\syit>java InterfaceExample  
Woof! Woof!
```

**B. Write a java code to implement a package concept.****Source Code:****File 1:**

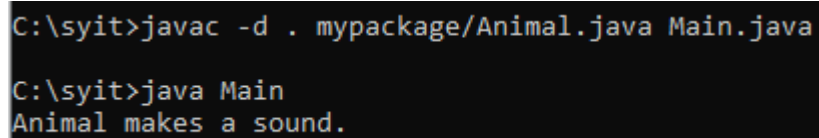
```
package mypackage;

public class Animal {
    public void makeSound() {
        System.out.println("Animal makes a sound.");
    }
}
```

**File 2:**

```
import mypackage.Animal;

public class Main {
    public static void main(String[] args) {
        Animal myAnimal = new Animal();
        myAnimal.makeSound();
    }
}
```

**Output:**

```
C:\syit>javac -d . mypackage/Animal.java Main.java
C:\syit>java Main
Animal makes a sound.
```

**C. Write a java code to implement a concept of exception handling.****Source Code:**

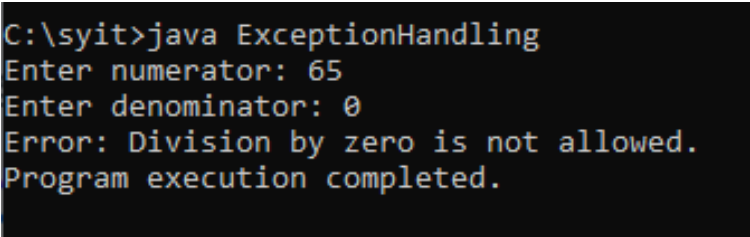
```
import java.util.Scanner;

public class ExceptionHandling {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        try {
            System.out.print("Enter numerator: ");
            int numerator = scanner.nextInt();

            System.out.print("Enter denominator: ");
            int denominator = scanner.nextInt();

            int result = numerator / denominator;
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            System.out.println("Error: Division by zero is not allowed.");
        } catch (Exception e) {
            System.out.println("Error: Invalid input. Please enter valid numbers.");
        } finally {
            scanner.close();
            System.out.println("Program execution completed.");
        }
    }
}
```

**Output:**

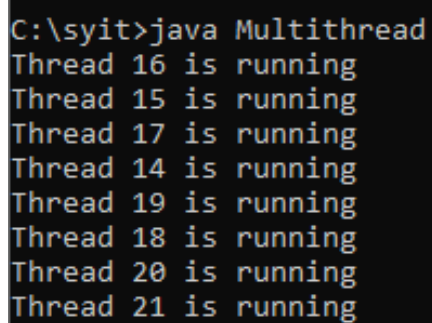
```
C:\syit>java ExceptionHandling
Enter numerator: 65
Enter denominator: 0
Error: Division by zero is not allowed.
Program execution completed.
```

**Practical - 8**

**A. Write a java code to design a class that demonstrates multi-threading.**

**Source Code:**

```
class Multithreading extends Thread {  
    public void run()  
    {  
        try {  
            System.out.println("Thread " + Thread.currentThread().getId() + " is running");  
        }  
        catch (Exception e) {  
            System.out.println("Exception is caught");  
        }  
    }  
}  
  
public class Multithread {  
    public static void main(String[] args)  
    {  
        int n = 8;  
        for (int i = 0; i < n; i++) {  
            Multithreading object = new Multithreading();  
            object.start();  
        }  
    }  
}
```

**Output:**

```
C:\syit>java Multithread  
Thread 16 is running  
Thread 15 is running  
Thread 17 is running  
Thread 14 is running  
Thread 19 is running  
Thread 18 is running  
Thread 20 is running  
Thread 21 is running
```

**Practical - 9****A. Write a java code to develop a SignIn form.****Source Code:**

```
import javax.swing.*;
import java.awt.event.*;

public class SignInForm {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Sign In");
        frame.setSize(300, 150);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        JLabel userLabel = new JLabel("Username:");
        userLabel.setBounds(20, 20, 80, 25);
        frame.add(userLabel);

        JTextField userField = new JTextField();
        userField.setBounds(100, 20, 160, 25);
        frame.add(userField);

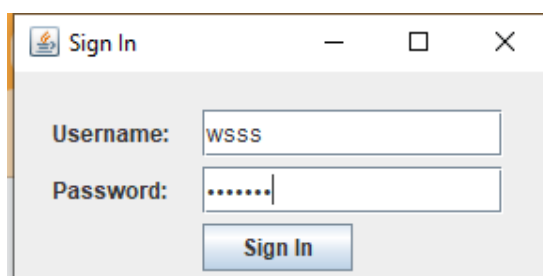
        JLabel passLabel = new JLabel("Password:");
        passLabel.setBounds(20, 50, 80, 25);
        frame.add(passLabel);

        JPasswordField passField = new JPasswordField();
        passField.setBounds(100, 50, 160, 25);
        frame.add(passField);

        JButton signInButton = new JButton("Sign In");
        signInButton.setBounds(100, 80, 80, 25);
        frame.add(signInButton);

        signInButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                JOptionPane.showMessageDialog(frame, "Sign In Clicked");
            }
        });

        frame.setVisible(true);
    }
}
```

**Output:**

**B. Write a java code to develop a hotel form.****Source Code:**

```
import javax.swing.*;
import java.awt.event.*;

public class HotelForm {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Hotel Booking Form");
        frame.setSize(350, 300);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        JLabel nameLabel = new JLabel("Name:");
        nameLabel.setBounds(20, 20, 100, 25);
        frame.add(nameLabel);
        JTextField nameField = new JTextField();
        nameField.setBounds(120, 20, 180, 25);
        frame.add(nameField);

        JLabel roomLabel = new JLabel("Room Type:");
        roomLabel.setBounds(20, 60, 100, 25);
        frame.add(roomLabel);
        String[] roomTypes = {"Single", "Double", "Suite"};
        JComboBox<String> roomBox = new JComboBox<>(roomTypes);
        roomBox.setBounds(120, 60, 180, 25);
        frame.add(roomBox);

        JLabel nightsLabel = new JLabel("Nights:");
        nightsLabel.setBounds(20, 100, 100, 25);
        frame.add(nightsLabel);
        JTextField nightsField = new JTextField();
        nightsField.setBounds(120, 100, 180, 25);
        frame.add(nightsField);

        JButton submitButton = new JButton("Book Now");
        submitButton.setBounds(120, 150, 100, 30);
        frame.add(submitButton);

        submitButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                JOptionPane.showMessageDialog(frame, "Booking Confirmed!\nName: " +
                    nameField.getText() + "\nRoom: " + roomBox.getSelectedItem() + "\nNights: " +
                    nightsField.getText());
            }
        });

        frame.setVisible(true);
    }
}
```



Output:

